

Deliverable 3: Capstone Project Analyzing/Data Processing Template

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Project Title: CAPSTONE: RETAIL ANALYTICS-PREDICTING CUSTOMER PURCHASE BEHAVIOR

Objectives:

- Perform your Analysis: Hypothesis
 - * (Curiosity got the best of me and I was able to come up with 5 hypotheses from the dataset)
- Hypothesis 1 (Income):
 - Customers with higher income are more likely to cater to the marketing campaigns as opposed to customers with lower incomes
- Null hypothesis 1:
 - There is no correlation/no significant relationship between customer income and campaign response
- Hypothesis 2 (education):
 - Customers with higher education (PhD, Master) are more likely to respond to the campaign than those with lower education
- Null Hypothesis 2:
 - Education level has no significant effect on a customer's behavior for responding to marketing campaign
- Hypothesis 3 (Age):
 - Younger customers ages 18-35 years old spend less on products compared to older age groups such as 46-55 years old.
- Null hypothesis 3:
 - There is no significant difference in spending based on customer age groups.

- Hypothesis 4 (Household):
 - Customers with children or teenagers at home spend less overall than customers without dependants.
- Null hypothesis 4:
 - There is no significant effect on customer spending based on whether they have children or teenagers at home.
- Hypothesis 5 (Purchase channel):
 - Customers who make more web purchases or catalog purchases are more likely to respond to the campaign
- Null Hypothesis 5:
 - There is no significant relationship with campaign response.
- What are you trying to find in the data?
 - The big question here is asking myself, What drives a customer to respond to a marketing campaign. There are a lot of factors, derived from the data, that can all ultimately have an impact on customer response to marketing campaigns, but my job is to use my curiosity to find what factor will end up driving the most impact towards customer behavior with marketing campaigns.
 - The dataset revolves around columns that have specific demographics such as customer income, age, and education. While also having columns that touch on the spending behavior of our customers such as total spend and purchasing behavior.
- Main Hypothesis:
 - There is a significant influence with the likelihood of a customer response to a retail marketing campaign based on customer demographics (income, education, age) and their spending behavior (total spend, purchase channel)
- Main Null Hypothesis: There is no significant effect on campaign response based on customer demographics and spending behavior.

- Perform your Analysis: What have you found from your Data/EDA (exploratory data analysis)
 - 2240 customers
 - 27 columns
 - Enrollment period: 2012-2014 (2 year range)
- Customer Demographics:
 - Age: Mean age is 57 years old. Age ranges from 30 yrs old to 130 yrs old (outlier born in 1893), with most customers being 40-60 yrs old.
 - Education: Most customers are Graduation-level. Followed by PHD and Master graduates. A small group has only basic education
 - Marital Status: Married, single, together, divorced, widow
 - Income: \$0 to \$666k, median \$51,382
- Spending Behavior
 - Wines - highest spend category
 - Meat products - second highest
 - Gold, Fish, sweets, fruits - lowest spend
- Campaign Performance
 - There are 5 campaigns (CMP1 - CMP5)
 - A final campaign (response column)
 - Overall response rate is 15%, indicating that most customers did not respond to the last campaign (response column)
 - Cmp acceptance rates vary, some outperformed others
 - Customers with higher income and education tend to respond more to the campaigns
- Purchase Channels
 - 4 channels: Web, Catalog, Store, and Deals
 - Store purchases are the most common channel
 - Web visits , approximately 5 per month (stats summary)
- Household composition
 - Average of 0.44 children and 0.51 teenagers per household
 - This lets me know that most customers have small or no families at home

- Key Note takeaways:
 - Wines and meat drive the most revenue - prime candidates for campaign promotions
 - Graduation level, middle aged, and higher income customers are the core groups that respond to marketing campaigns most.
 - Campaign response is low- 15%. Lets company aware that there is room for improvement for targeting
 - Stores are the most common purchasing channel while digital channels (web/catalog) are second.
 - Household size has little to no significance- indicates that campaigns aimed at families may underperform.
- Data Visuals
 - Tableau charts - utilizing claude for inspiration on data visualization templates
- SQL
 - Findings
 - Queries
 - Notes